

AN EFFICIENT FRAMEWORK FOR LONG-TAILED AND MULTI-LABEL CLASSIFICATION ON CHEST X-RAYS

Huy Le Pham^{*} Khoa Ha Anh^{*} Nguyen-Thanh-Thao Vo^{*} Ky Trung Nguyen^{*†}
 Chau Ly Thi Huyen^{**} Hien Q. Ta^{*†} Thang Doan Van⁺⁺

^{*} School of Computer Science and Engineering, International University, Vietnam

[†] Vietnam National University, Ho Chi Minh City, Vietnam

^{**} Faculty of Information Technology, Van Lang School of Technology

Van Lang University, Ho Chi Minh City, Vietnam

⁺⁺ Faculty of Information Technology, Industrial University of Ho Chi Minh City,
 Ho Chi Minh City, Vietnam

ABSTRACT

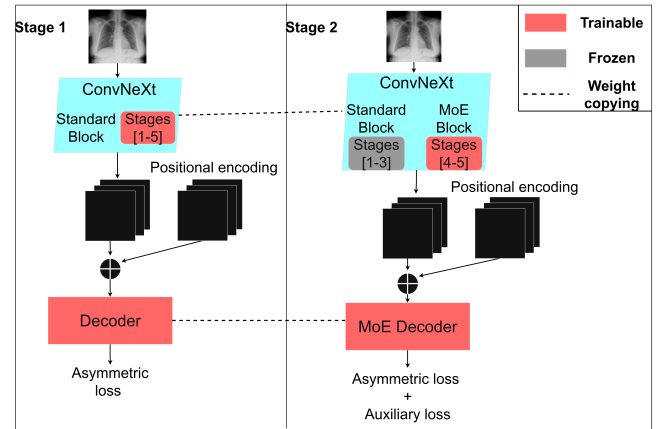
To tackle the challenges of long-tailed disease distributions and label-occurrence within chest X-ray classification. This paper introduces our solution, a framework of two stages: (1) ConvNeXt and query-based decoder approach; (2) with a specialised diverse embedding fusion module, MoE ConvNeXt. This module utilizes attention mechanisms to intelligently integrate features across images and address label correlations. Combined with two-stage training strategy and MoE integration, our solution achieved top performance, obtaining an mAP of 0.4599 on the CXR-LXT test set and securing third place in the ISBI 2026 Task 1.

Index Terms— Medical Image Classification, Long-Tail Distribution, Label-occurrence, ConvNeXt, Mixture of Experts.

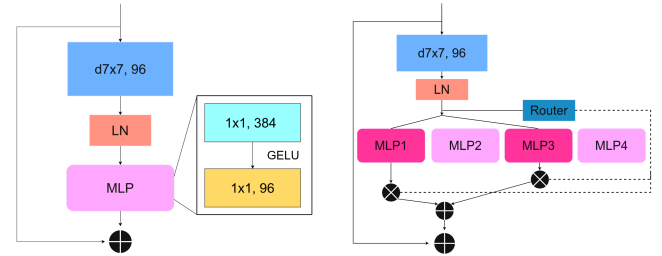
1. INTRODUCTION

Disease recognition from medical images is becoming increasingly necessary in decision-making systems, supporting doctors in making more accurate diagnoses thanks to the automatic, rapid, and accurate recognition capabilities of machine learning models. However, building a lightweight machine learning model that not only predicts accurately and quickly but also does not require excessive GPU resources to train on large datasets remains a challenge. Moreover, one advantage of this approach is that it does not require overly large models like Vision Transformers (ViT) [1], which often require many GPU hours to train on large datasets. Furthermore, building and training deep learning machine learning models for multi-disease recognition in medical images often suffer from problems such as: (1) long tail distribution; (2) label co-occurrence.

Therefore, in this paper, we develop a framework that enables the use of a lightweight deep learning model while



(a) Two-stages of the framework



(b) Standard Block

(c) MoE Block

Fig. 1. Our proposed framework (a) and its detailed blocks (b) and (c) .

maintaining its performance for multi-disease recognition from medical images. Additionally, we propose several approaches to address label co-occurrence imbalance, thereby improving the accuracy of the machine learning model. The framework, illustrated in Fig. 1, has two stages.

In the first stage, medical images are processed and aug-

mented using methods such as rotation, blur, etc. Then, the augmented medical images are extracted into visual embeddings using ConvNeXt, after which they are fed into an introduced decoding head to produce the final output probabilities for each multi-label. In the second stage, after training the model using the method described in stage 1, the model’s checkpoints will be saved, and the model parameters in stage 1-3 will be frozen. It is worth noting that the model is only tuned in stages 4-5, in combination with Mixture of Experts (MoE), which not only preserves the general representation learned in early layers but also enabling expert specialization in high-level visual features and decoder representations. This helps the model learn and generalize better, capable of increasing the accuracy of disease classification on unseen data.

In summary, with this approach, we have the following main contributions:

- Stage 1: Building a ConvNeXt model combined with a query-based decoder to solve the problem of multi-disease classification from medical images.
- Stage 2: Incorporate Mixture-of-Experts into the stage-1 model to introduce conditional computing and disease specialization. Through this sparse MoE mechanism, the model can learn diversified feature representations and implicitly capture some of the disease-related pattern, thereby further improve its performance on unseen data.

The remainder of this paper is presented in Section 2, where we describe our methodology. In Section 3, we present the results from the experiment. Eventually, our work is concluded in Section 4.

2. PROPOSED METHOD

2.1. Data preprocessing

We follow the classic pipeline consisting of image resizing, data augmentation and intensity normalization. During training, a plenty of data augmentation techniques are applied on-the-fly to increase the data diversity, including random resized cropping, horizontal flipping, affine transformations (random shifts, scaling, and rotations), elastic and grid-based spatial distortions, random brightness and contrast adjustments, as well as noise and blur operations such as Gaussian noise, Gaussian blur, motion blur, and median blur. These transformations are applied probabilistically to simulate realistic variations in acquisition conditions and anatomical appearance while preserving diagnostic semantics. Finally, the data is normalized following the transformation as used for training under ImageNet [2].

2.2. Model architecture

Our main architecture involves a ConvNeXt [3] network and a Query-based decoding head for classification. Regarding ConvNeXt, it follows a four-stage hierarchy design, where the input image is progressively processed through

a stack of standard ConvNeXt block, demonstrated in Fig 1.b, and spatial downsampling. Initially, the medical image $I \in R^{H \times W \times C}$ is mapped into a high-level latent representation $F \in R^{H' \times W' \times D}$ through this image backbone. Moreover, inspired by [4, 5, 6], we adopt a query-based decoding head as a replacement for the global average pooling (GAP) [7] for multi-label prediction. The decoder is initialized with N learnable query embeddings $Q_0 \in R^{N \times D}$, where N is the number of target labels. It processes these queries through a stack of cross-attention and self-attention layers followed by a classic Feedforward transformation (FFN), which is a little bit different compared to [5, 6]. We want to let these queries interact with the visual tokens via cross-attention first before refining itself through the self-attention mechanism. The final decoded queries $Q \in R^{N \times D}$ are then projected to output logits $y \in R^N$. Additionally, we follow DETR [8] to add a two-dimensional positional encoding $P \in R^{H' \times W' \times D}$ on top of the extracted feature map from the ConvNeXt backbone to inject the absolute spatial information before interacting it with queries from the decoder head.

2.3. Mixture of experts

As we seek further disease specialization beyond a single shared representation, we introduce a sparse Mixture-of-Experts (MoE) mechanism to enable conditional computation within the model. Given an intermediate representation $x \in R^{T \times D}$, where T denotes the number of tokens and D denotes the feature dimension, a lightweight router predicts routing scores over E experts and selects the top- k experts per token. Each token is processed only by its selected experts, and their outputs are combined using normalized routing weights. Specifically, we implement the MoE following the Gshard [9] design style.

To avoid expert collapse and encourage balanced expert utilization, an auxiliary load-balancing loss is introduced during training. Let $G \in R^{T \times E}$ denote the sparse gating matrix, where T is the number of tokens and E is the number of experts. Each element $G_{t,e}$ represents the normalized routing weight assigned from token t to expert e (non-zero only for the selected top- k experts). The importance and load of expert e are defined as

$$Imp_e = \sum_{t=1}^T G_{t,e}, \quad Load_e = \sum_{t=1}^T \mathbf{1}(G_{t,e} > 0), \quad (1)$$

where Imp_e measures the total routing weight assigned to expert e , and $Load_e$ counts the number of tokens routed to it. The auxiliary loss is then formulated as

$$L_{aux} = \frac{E}{T^2} \sum_{e=1}^E Imp_e \cdot Load_e, \quad (2)$$

2.4. Training

The model is optimized using a two-stage training strategy, which is shown in Fig 1.a. In the first stage, the base architecture—comprising the convolutional backbone and the query-based decoder—is trained end-to-end with the primary classification objective. This stage aims to learn shared and robust visual representations that capture common pathological patterns across all disease categories.

In the second stage, MoE modules are introduced to MLP components of the ConvNeXt block in stage 4+5 and FFN layers of the decoding head. We first initialize the model using the pre-trained weights obtained in the first training stage and freeze most of the backbone parameters. In other words, we initialize each expert weights from the base layer and add a small noise to each of them for different specialization. During training, only parameters from the ConvNeXt block in stage 4+5 and the decoder are activated. Moreover, at this training stage, the total training objective consists of the primary classification loss augmented with a lightly weighted auxiliary load-balancing loss to encourage diverse expert utilization.

2.5. Loss Function

To address class imbalance inherent in multi-label medical image classification, we adopt the Asymmetric Loss [10] as the primary training objective. Given the predicted logits $y \in R^N$ and the corresponding primary ground-truth labels $t \in \{0, 1\}^N$, the loss applies asymmetric focusing factors to positive and negative samples, reducing the contribution of easy negatives while maintaining sensitivity to positive labels. In addition to the classification loss, when MoE modules are adopted, an auxiliary load-balancing loss is added to encourage uniform expert utilization. The final training objective then is given by $L = L_{asym} + \lambda * L_{aux}$, where λ is a small weighting coefficient controlling the influence of the auxiliary loss.

3. IMPLEMENTATION AND RESULTS

3.1. Dataset

The dataset is provided with high-quality benchmark CXR data, which used in this challenge is based on a curated set of over 160,000 chest X-ray images selected from the PadChest [11], supplemented with PadChest-GR [12], where radiologists provided manual annotations. After that, the organizer curated a final set of 36 long-tailed classes, which span a broad range of chest findings and serve as the prediction targets for ISBI 2026 Task 1. The details of data splitting, construction methods with the evaluation metrics can be found in [13]. Specifically, the task are directly motivated by real-world diagnostic scenarios, where disease distributions are highly imbalanced, and ground truth labels may vary in

quality and coverage. By addressing these challenges, CXR-LT 2026 encourages the development of methods can better handle data imbalance and label uncertainty in clinical practice.

3.2. Implementation details

Regarding the main backbone, we use the ConvNeXt-B architecture pre-trained from ImageNet. All input images are resized to a fixed spatial resolution of 384x384 during training and evaluation. For the MoE configuration, we use top-k routing with $k = 2$ and the number of experts as 4. We use Adam optimizer with the initial LR of $4 * 10^{-5}$ and weight decay of 0.01. Moreover, the auxiliary load-balancing loss is weighted by $\lambda = 5 * 10^{-3}$. The training process follows the two-stage strategy described in Section 2.4. In the first stage, the base model is trained end-to-end for 10 epochs. In the second stage, MoE modules are introduced and trained for an additional 2 epochs while most backbone parameters remain frozen. All experiments are conducted on a single NVIDIA Tesla P40 GPU.

3.3. Results

For task 1, we decided to divide the existing training samples into 103,000 samples for model training and 17,000 samples for internal validation. This validation split is used exclusively for monitoring performance and selecting checkpoints during training and is not part of the official challenge evaluation. Table 1 reports the performance of the proposed model under the two-stage training strategy on the internal validation set.

Table 1. Model performance on the internal validation set in two stages.

Stage	mAP	mF1	mAUC	mECE
1	0.385	0.395	0.875	0.134
2	0.405	0.402	0.882	0.130

On the other hand, following the internal evaluation, we are able to achieve an mAP of 0.505 on the Development set with the Stage-1 model checkpoint, where the mAP is around 0.385. This result on the Development set is demonstrated in Table 2.

Table 2. Stage-1 model performance on the Development set.

Stage	mAP	mF1	mAUC	mECE
1	0.505	0.415	0.874	0.776

Finally, the checkpoint from stage 2 was used to submit to Task 1 of the challenge, achieving third place with the following score on the final test set (Table 3).

Table 3. Model performance on the Test set.

Stage	mAP	mF1	mAUC	mECE
2	0.4599	0.4504	0.8827	N/A

4. CONCLUSION

In this work, we conducted experiments with ConvNeXt-based approach featuring a query-based decoder to address the long-tail distribution in multi-label recognition from Chest X-Ray images. In fact, the Query2Label method shows that it can improve ConvNeXt’s performance via its query mechanism. Moreover, thanks to the training strategy with Mixture of Experts’ capability, the learning model can generalize even better, leading to better recognition performance on unseen data.

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